

- 3 A train initially at rest accelerates at a constant rate of 1.4 m s^{-2} for 10 s and then slows down at a constant rate of 0.20 m s^{-2} until it comes to a rest. The total distance travelled by the train is

A 80 m

B 220 m

C 560 m

D 1120 m

$$s_1 = u_1 t_1 + \frac{1}{2} a_1 t_1^2 = \frac{1}{2} \times 1.4 \times 10^2 = 70 \text{ m}$$

$$v_1 = u_1 + a_1 t_1 = 1.4 \times 10 = 14 = u_2$$

$$v_2^2 = u_2^2 + 2a_2 s_2$$

$$s_2 = \frac{u_2^2}{2a_2} = \frac{14^2}{2 \times 0.2} = 490 \text{ m}$$

$$s = s_1 + s_2 = 70 + 490 = 560 \text{ m}$$

- 4 A falling stone strikes some soft ground at speed v and suffers a constant deceleration until it